Supplementary Table 4 | Quantitative ablation study of individual modules in the CELLECT framework. We assessed the contributions of the Center Enhancement Network (CEN), intra-frame MLP, and inter-frame MLP by progressively enabling each. Tracking Accuracy is defined as the proportion of uninterrupted, error-free trajectories over 50 to 200 frames[2] and Error Links refer to the total tracking errors (false negatives, false negative divisions, and identity switches). In ablated models, nearest-neighbor matching replaced the inter-frame MLP, and redundancy filtering was disabled without the intra-frame MLP. Removing CEN resulted in noisy center proposals due to unenhanced confidence maps. Model was trained on mskcc-confocal #3 and evaluated #2.

3D U-Net	CEN	Intra-frame MLP	Inter-frame MLP	Tracking Accuracy[2] Length (frames)				Error Links (FN+FND+IS)
				50	100	150	200	
Yes	No	No	No	0.76	0.42	0.13	0.02	382
Yes	Yes	No	No	0.84	0.55	0.21	0.10	219
Yes	Yes	Yes	No	0.85	0.57	0.23	0.10	210
Yes	Yes	Yes	Yes	0.94	0.84	0.61	0.32	138